

Thought 1.

contrapositive (converse ($p(x) \rightarrow q(x)$))

$$= \underline{\text{inverse}} (\neg x (p(x) \rightarrow q(x)))$$

Prob 1:

Truth tables.

Draw TT: $\{ \begin{array}{l} p \rightarrow (q \rightarrow r) \\ (p \wedge q) \rightarrow r \end{array} \}$

→ Comment on your observation.

			$p \rightarrow x$		$y \rightarrow r$	
p	q	r	$p \rightarrow (q \rightarrow r)$	$p \wedge q$	$p \rightarrow x$	$y \rightarrow r$
					0	
					1	X
					0	
					1	
					0	
					1	
					0	
					1	

Logically equivalent!

$$\begin{array}{c}
 (u \rightarrow r) \wedge \\
 ((r \wedge s) \rightarrow (p \vee t)) \wedge \\
 (q \rightarrow (u \wedge s)) \wedge \\
 (\neg t) \rightarrow
 \end{array}
 \quad \left. \vphantom{\begin{array}{c} (u \rightarrow r) \wedge \\ ((r \wedge s) \rightarrow (p \vee t)) \wedge \\ (q \rightarrow (u \wedge s)) \wedge \\ (\neg t) \rightarrow \end{array}} \right\}$$

$$\therefore (q \rightarrow p)$$

$$\begin{array}{c}
 (u \rightarrow r) \wedge \\
 ((r \wedge s) \rightarrow (p \vee t)) \wedge \\
 (q \rightarrow (u \wedge s)) \wedge \\
 (\neg t) \wedge \\
 (q) \rightarrow
 \end{array}
 \quad \left. \vphantom{\begin{array}{c} (u \rightarrow r) \wedge \\ ((r \wedge s) \rightarrow (p \vee t)) \wedge \\ (q \rightarrow (u \wedge s)) \wedge \\ (\neg t) \wedge \\ (q) \rightarrow \end{array}} \right\}$$

$$\therefore p$$

$$r(x): 2x+1=5$$

$$s(x): x^2=9$$

$$\exists x r(x) \wedge \exists x s(x) \rightarrow \text{true.}$$

$$\exists x [r(x) \wedge s(x)] \rightarrow \text{false.}$$

①

$$\boxed{\exists x [r(x) \wedge s(x)] \not\Rightarrow \exists x r(x) \vee \exists x s(x)}$$

②

$$\exists x [r(x) \wedge s(x)] \not\Rightarrow \exists x r(x) \wedge \exists x s(x)$$

③

$$\exists x r(x) \vee \exists x s(x) \not\Rightarrow \exists x [r(x) \wedge s(x)]$$

$$\exists x [p(x) \wedge q(x)] \Rightarrow [\exists x p(x) \wedge \exists x q(x)]$$

$$\exists x [p(x) \vee q(x)] \Leftrightarrow [\exists x p(x) \vee \exists x q(x)]$$

$$\forall x [p(x) \wedge q(x)] \Leftrightarrow [\forall x p(x) \wedge \forall x q(x)]$$

$$[\forall x p(x) \vee \forall x q(x)] \Rightarrow \forall x [p(x) \vee q(x)]$$

$$\neg [\forall x p(x)].$$

"It is not the case that for all x , $p(x)$ is true"

"there ~~must~~ exist an x for which $\neg p(x)$ is true"

$$\neg [\forall x p(x)] \Leftrightarrow \exists x \neg p(x).$$

$\exists x p(x)$ is true (false)

$\neg \exists x p(x) \rightarrow \text{false (true)}$

$$\neg (\exists x p(x)) \Leftrightarrow \forall x \neg p(x).$$

- ① $\neg [\forall x \overbrace{p(x)}^{\neg p(x)}] \Leftrightarrow \exists x \neg p(x)$
- ② $\neg [\exists x \overbrace{p(x)}^{\neg p(x)}] \Leftrightarrow \forall x \neg p(x)$
- ③ $\neg [\forall x \neg p(x)] \Leftrightarrow \exists x \neg \neg p(x) \Leftrightarrow \exists x p(x)$
- ④ $\neg [\exists x \neg p(x)] \Leftrightarrow \forall x \neg \neg p(x) \Leftrightarrow \forall x p(x)$

Rules of negation.

E.g. 1:

$p(x)$: x is odd.

$q(x)$: $x^2 - 1$ is even.

$$\neg [\forall x (p(x) \rightarrow q(x))] \Leftrightarrow \exists x [\neg (p(x) \rightarrow q(x))].$$

Shorthand:

$$\forall x \forall y \forall z [x + (y + z) = (x + y) + z]$$

$$\forall y \forall x \forall z [x + (y + z) = (x + y) + z]$$

$$\forall x, y, z p(x, y, z) \Leftrightarrow \forall x, y, z p(x, y, z)$$

3! ways.

$$\Leftrightarrow \forall z, x, y p(x, y, z) \dots$$

E.g. 2.

$$\neg (\forall x \exists y [(p(x,y) \wedge q(x,y)) \rightarrow r(x,y)])$$



~~$$\exists x \forall y [(p(x,y) \wedge q(x,y)) \rightarrow r(x,y)]$$~~

$$\neg [\exists x \forall y [(p(x,y) \wedge q(x,y)) \wedge \neg r(x,y)]]$$

$$\neg (\forall x \exists y [(p(x,y) \wedge q(x,y)) \rightarrow r(x,y)])$$

$$\exists x \neg \exists y [(p(x,y) \wedge q(x,y)) \rightarrow r(x,y)]$$

$$\exists x \forall y \dots$$

Rule: Rule of universal specification.

if $\forall x p(x)$ is true, then $p(a)$ is true
for all 'a' in the universe.

$m(x)$: x is a maths prof
 $c(x)$: x has studied calculus.

All maths prof have studied calculus $\rightarrow$ generic $(\forall x)$

specific person $\leftarrow$ [Leona is a maths prof.
Therefore Leona has studied calculus.
(specific x)

$\forall x [m(x) \rightarrow c(x)] \rightarrow$

$\frac{m(l)}{\therefore c(l)}$

$$1. \quad \forall x [m(x) \rightarrow c(x)]$$

$$2. \quad m(l)$$

$$3. \quad m(l) \rightarrow c(l)$$

[Rule of univ spec].

$\therefore c(l)$ $\longrightarrow$ Modus Ponens.

E.g. 3.

$p(t)$: t has two sides of equal length

$q(t)$: t is an isosceles triangle.

$r(t)$: t has two angles of equal measure.

$\triangle XYZ \longrightarrow 'c'$

(a) In triangle XYZ there is no pair of angles of equal measure. $\neg r(c)$

(b) If a triangle has two sides of equal length then it is isosceles. $\forall t [p(t) \rightarrow q(t)]$

(c) If a triangle is isosceles then it has two angles of equal measure $\forall t [q(t) \rightarrow r(t)]$.

(d) Therefore $\triangle XYZ$ has no two sides of equal measure. $\therefore \neg p(c)$

E.g. 4

$j(x)$: x is a junior

$s(x)$: x is a senior

$p(x)$: x is enrolled in a physical education class.

→ (a) ^{any} No junior or senior ^{not} is enrolled in a physical education class.

(b) Mary is enrolled in a physical education class.

(c) Therefore Mary is not a senior.

Frame the stubs & prove the validity of the arguments.

$$\forall x [(j(x) \vee s(x)) \rightarrow \neg p(x)]$$

$p(m)$

$$\therefore \neg s(m)$$

$$(1) \quad \forall x \neg [(j(x) \vee s(x)) \rightarrow \neg p(x)]$$

$$(2) \quad p(m)$$

$$(3) \quad [(j(m) \vee s(m)) \rightarrow \neg p(m)]$$

rule of
specification

$$(4) \quad p(m) \rightarrow \neg (j(m) \vee s(m))$$

$$p \rightarrow q$$

$$\Leftrightarrow$$

$$\neg q \rightarrow \neg p$$

$$p(m) \rightarrow \neg j(m) \wedge \neg s(m)$$

$$(5)$$

$$\neg j(m) \wedge \neg s(m)$$

$$(6)$$

$$\neg s(m)$$

Rule of
conjunction

Rule of universal generalization.

x is replaced by any a from the universe $\Rightarrow p(x)$ is true.
 $p(x)$ is true for ~~any~~ ^{all} arbitrary a in the universe then $\forall x p(x)$ should be true.

E.g. 5.

$$\forall x [p(x) \rightarrow q(x)]$$

$$\forall x [q(x) \rightarrow r(x)]$$

$$\therefore \forall x [p(x) \rightarrow r(x)]$$

① $\forall x [p(x) \rightarrow q(x)] \rightarrow$ premise.
 $\rightarrow$ R. of spec.

② $p(c) \rightarrow q(c)$

③ $\forall x [q(x) \rightarrow r(x)] \rightarrow$ premise
 $q(c) \rightarrow r(c) \rightarrow$ R. of spec

④

$p(c) \rightarrow r(c) \rightarrow$ Syllogism.

⑤

⑥ $\forall x [p(x) \rightarrow r(x)] \rightarrow$ R. of gen.

E.g. $\forall x [p(x) \vee q(x)]$
 $\forall x [(\neg p(x) \wedge q(x)) \rightarrow r(x)]$

 $\therefore \forall x [\neg r(x) \rightarrow p(x)]$

① $\forall x [p(x) \vee q(x)]$

② $p(a) \vee q(a)$

③ $\forall x [(\neg p(x) \wedge q(x)) \rightarrow r(x)]$

④ $(\neg p(a) \wedge q(a)) \rightarrow r(a)$

⑤ $\neg r(a) \rightarrow \neg (\neg p(a) \wedge q(a))$

⑥ $\neg r(a) \rightarrow (p(a) \vee \neg q(a))$

⑦ $\neg r(a) \rightarrow \text{Assume.}$

⑧ $p(a) \vee \neg q(a)$

⑨ $p(a) \vee [q(a) \wedge \neg q(a)]$

⑩ $p(a)$

 $\forall x [\neg r(x) \rightarrow p(x)]$

Defⁿ:

Let n be an integer.

We call n to be even if n is divisible by 2.

$\exists r$ which is an integer
s.t. $n = 2r$.

If n is not even, then we call it odd

$\exists s$, s.t. $n = 2s + 1$.

Theorem 1.

For all integers k & l , if both k & l are odd then $k+l$ must be even.

$$k = 2a + 1$$

$$l = 2b + 1$$

$$k+l = 2a + 2b + 2 = 2(a+b+1)$$

$\hookrightarrow \therefore k+l$ is even

5.

the
5/25

Theorem

E.g. 6.

Prove or disprove:

If n is an integer then
 $n^2 = n$ or $\neg \forall n [n^2 = n]$.

Counter example.

$$n=0, n=1.$$

$$\exists n [n^2 = n]$$

but
 ~~$\forall n [n^2 = n]$~~

Theorem 2:

For all integers k & l if k and l are odd
then kl is also odd.

$$k = 2a + 1$$

$$l = 2b + 1$$

$$kl = (2a + 1)(2b + 1)$$

$$= 4ab + 2a + 2b + 1$$

$$= 2(\underbrace{2ab + a + b}_{\text{sum is an integer}}) + 1$$

$$= 2s + 1 \quad \text{hence odd.}$$

Theorem 3: If m is an even integer,
 $m+7$ is odd.

$m=2a$ a is an integer.

$$m+7 = 2a+7$$

$$= 2a+6+1$$

$$= 2(\underbrace{a+3})+1$$

$\hookrightarrow$ integer.

$$= 2s+1 \quad \text{proved.}$$

Theorem 4:

For all positive ^{real} integers x & y if the
product xy exceeds 25 then $x > 5$ or $y > 5$.

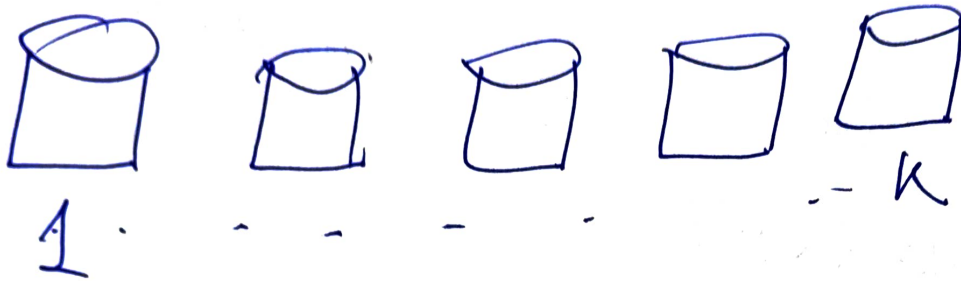
Let us assume ~~that~~ $0 < x \leq 5$ &

$$0 < y \leq 5$$

$$\Rightarrow 0 < x \cdot y \leq 25$$

$\Rightarrow$ product does not exceed 25
 $\therefore$ Either x or y has to be > 5 for the
product $xy > 25$

Pigeons & holes.



n pigeons to be accommodated.

$$n > k.$$

↳ Observation:

At least some pigeons will need to share holes.

↳ pigeon-hole principle.